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Software Requirements Specification

SWE 4404: SPL-II

Project Title: Dengue Alert & Ward-Based Management System

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the **Dengue Alert & Ward-Based Management System**, a mobile-first Progressive Web App (PWA) that lets ordinary citizens report possible dengue breeding spots and lets city authorities act on those reports quickly and accountably.

Citizens report a hazard with a live photo and an automatically captured GPS location. The system routes the report to the Ward Councillor responsible for that location, the Councillor assigns a Ward Worker to clean it up, and the Worker submits a geo-tagged proof photo. The Councillor compares the before and after photos before closing the case. Areas that collect a significant number of complaints are flagged as hotspots and escalated to the Zonal Executive Magistrate.

1.2 Document Conventions

- Epics are identified as **EP-x**.
- User stories are identified as **US-x.y**, where x is the epic number.
- Priority follows the MoSCoW classification below.

Priority	Meaning
Must	Essential feature; the system is not acceptable without it
Should	Important, but can wait if time is short
Could	Optional; implemented only if time permits
Won't	Not included in the current release

1.3 Intended Audience

Audience	Purpose
Citizens	Report dengue hazards and follow the progress of their complaints
Ward Councillors	Review, assign and close complaints in their own ward
Ward Workers / Inspectors	Carry out cleanup tasks and upload proof of completion
Super Admin / System Analyst	Manage users, wards, thresholds and city-wide reports
Zonal Executive Magistrate	View hotspot reports through a secure, login-free link
Developers and Testers	Implement and verify the requirements
Supervisor	Evaluate the project against the stated scope

1.4 Product Scope

The product aims to close the gap between public complaints and the city's cleanup crews, so that dengue hazards are dealt with early and every action can be traced. Asking people to install a heavy native app discourages reporting, so the system runs in the browser and can be used instantly. The objectives are:

- Let any citizen report a hazard in a few taps, with a photo and an automatic location.
- Route each complaint to the correct Ward Councillor without manual address entry.
- Create accountability through before and after photo verification.

- Automatically escalate high-complaint areas to the Zonal Executive Magistrate.
- Warn citizens who physically enter a 200-metre radius of a high-risk hotspot.
- Support Bangla and English throughout.

1.5 Changes from the Original Proposal

The following decisions were made after the proposal and supervisor feedback. They override the proposal wherever the two differ.

#	Area	Original Proposal	Current Decision	Reason
1	Login and registration	Name plus phone number or email	Email only	Suggested by the supervisor; simpler and avoids SMS verification cost
2	Media storage	Firebase Cloud Storage	Cloudinary	Cloud provider asked for payment; Cloudinary free tier is sufficient
3	Magistrate escalation	48-hour link sent by Twilio / SSL Wireless SMS	48-hour link sent by email	SMS gateways carry a cost; email was the original backup plan
4	Complaint limit	Unlimited submissions	Maximum 3 complaints per user per 24 hours	Feedback: users could upload fake or unnecessary photos

2. Overall Description

2.1 Product Perspective

The system is a standalone, web-based platform with a React PWA front end and a Node.js (Express) REST back end. Spatial data is stored in PostgreSQL with the PostGIS extension, and images are stored on Cloudinary. The main parts are:

- **Citizen interface** (public landing view, reporting, complaint tracker)
- **Worker interface** (assigned tasks, map, proof upload)
- **Councillor dashboard** (ward complaints, assignment, verification)
- **Super Admin dashboard** (users, wards, thresholds, analytics)
- **Hotspot Escalation Module** (threshold check, secure link, email)
- **Spatial Database** (PostgreSQL + PostGIS)
- **External services** (Cloudinary, Mapbox / Google Maps, Email service, browser Push API)

2.2 Product Functions

- Guest overview, email registration and secure login
- Live photo capture with automatic GPS location
- Offline queuing and automatic resubmission of complaints
- Geofenced routing of complaints to the correct ward
- Duplicate grouping and priority sorting for Councillors
- Task assignment and tracking for Workers
- Geo-tagged, timestamped proof of completion and before/after verification
- Hotspot detection and email escalation with a 48-hour secure link
- Proximity push notifications within 200 m of a hotspot
- Complaint status tracker for citizens
- Bangla and English interface
- Administration, reporting and audit logging

2.3 User Classes and Personas

Access is controlled through a four-tier Role-Based Access Control (RBAC) model. The Magistrate and system services are external actors and do not hold an account.

User Class	Description	Account
Citizen	Reports hazards, tracks complaints, receives hotspot alerts. May browse the app before registering.	Yes (Tier 1)
Worker / Conservancy Inspector	Receives cleanup tasks, travels to the site, marks work started, uploads the After photo.	Yes (Tier 2)
Local Admin / Ward Councillor	Sees only own-ward reports, assigns Workers, verifies and closes cases.	Yes (Tier 3)

User Class	Description	Account
Super Admin / System Analyst (City Corporation)	Manages all users, wards, thresholds and views city-wide data.	Yes (Tier 4)
Zonal Executive Magistrate	Opens a time-limited hotspot report from an emailed link.	No (token link)
Email Service / Cloudinary / Maps	External services used by the system.	External

2.4 Complaint Status Lifecycle

Status	Meaning	Changed by
Submitted	Complaint received with photo and GPS location	System
Under Review	Routed to the ward and visible to the Councillor	System
Assigned	Councillor has verified the photo and assigned a Worker	Councillor
In Progress	Worker has marked Start Work on arrival	Worker
Awaiting Verification	Worker has uploaded the After photo	Worker
Closed	Councillor approved the before/after comparison	Councillor
Rejected	Councillor found the report invalid (for example, a fake or unrelated photo)	Councillor

2.5 Operating Environment

- Modern mobile and desktop browsers (Chrome, Firefox, Edge, Safari) with PWA support
- Mobile-first responsive interface
- Devices with a camera and GPS for citizens and workers
- Node.js server on Linux or Windows
- PostgreSQL with PostGIS extension

2.6 Constraints

- Location data must come from the device's GPS (HTML5 Geolocation); manual address entry is not offered.
- Photos must be captured live from the camera and must not be saved to the user's phone gallery.
- Each user may submit at most three complaints within any 24-hour period.
- The project must run on free-tier services (Cloudinary, email service, map provider) within the course timeline.
- Citizens must be able to see the app's overview before being asked to log in.
- Internet is needed for synchronisation, although reporting and worker actions must tolerate a weak connection.

2.7 Assumptions and Dependencies

- Ward boundary polygons are available and can be loaded into PostGIS.
- Users grant camera, location and notification permissions on their device.

- The Email service delivers messages reliably; the Magistrate's email address is configured per zone by a Super Admin.
- Cloudinary and the map provider remain available within their free usage limits.
- Councillors and Workers are registered by the Super Admin before the system goes live.

3. External Interface Requirements

3.1 User Interface

Interface	Main Screens
Citizen	Landing and app overview, Register / Login, Report screen (camera, auto location, optional description), My Complaints tracker, Hotspot alert, Language toggle
Worker	Assigned task list, Task detail (location, citizen photo), Map and directions, Start Work, Proof upload, Completed task history
Councillor	Ward dashboard (priority-sorted, grouped duplicates), Complaint detail, Assign Worker, Before/After comparison, Close or reject case, Hotspot alerts
Super Admin	User and role management, Ward management, Threshold and zone settings, City-wide reports, Audit logs
Magistrate (public link)	Read-only hotspot report page, valid for 48 hours, no login

3.2 Hardware Interface

- Device camera for live photo capture
- GPS receiver for location
- Network connection (Wi-Fi or mobile data)

3.3 Software Interface

Interface	Purpose
PostgreSQL + PostGIS	Store application data and perform ward-boundary and radius queries
Cloudinary	Store and deliver complaint (Before) and proof (After) images
Mapbox / Google Maps	Display complaint locations, hotspots and worker directions
Email service (Gmail-based)	Send registration, password reset, status and Magistrate escalation emails
Browser Push / Service Worker	Hotspot proximity notifications and offline queue

3.4 Communication Interface

- Client-server communication over HTTPS only.
- REST API using JSON between the React PWA and the Node.js back end.
- Escalation links contain a signed, random token that expires after 48 hours.

4. Epics and User Stories

EP-1: Authentication, Access Control and Profile Management

Epic Goal: Allow citizens, workers, councillors and administrators to access the system securely using only their email address.

Business Value: Keeps registration quick for citizens, lowers cost by avoiding SMS verification, and ensures each role sees only what it should.

Priority: Must

User Stories

ID	User Story	Priority
US-1.1	As a citizen, I want to see the app overview and functions before logging in, so that I can decide whether to use it.	Must
US-1.2	As a citizen, I want to register with only my name and email, so that signing up is fast.	Must
US-1.3	As a user, I want to log in with my email and password, so that my account stays protected.	Must
US-1.4	As a user, I want to reset my password through an email link, so that I can recover my account.	Should
US-1.5	As a user, I want to switch between Bangla and English, so that I can use the app in my own language.	Must
US-1.6	As a Super Admin, I want to create Councillor and Worker accounts and assign their ward, so that only authorised staff get those roles.	Must
US-1.7	As a user, I want to update my profile name, so that my details stay correct.	Should

Acceptance Criteria

Given	When	Then
A visitor opens the app without an account	The landing page loads	An overview and feature summary is shown without asking for login
A new citizen enters a valid name and email	Registration is submitted	An account is created and a verification email is sent
An email is already registered	Someone registers with the same email	The system rejects it and shows a clear message
A user enters correct credentials	Login is submitted	The user lands on the dashboard for their role
A user enters wrong credentials	Login is submitted	Access is denied and an error message is shown
A registered email is entered on the reset page	Reset is requested	A time-limited reset link is emailed
A user selects Bangla or English	The language is changed	All labels and messages switch immediately and the choice is remembered
A Super Admin creates a Councillor account	A ward is assigned	The Councillor can see only that ward's complaints

EP-2: Citizen Complaint Reporting and Tracking

Epic Goal: Let citizens report a dengue hotspot in a few taps, with proof and an exact location, and follow what happens next.

Business Value: More and better reports reach the authorities, and citizens see that their complaint is actually handled.

Priority: Must

User Stories

ID	User Story	Priority
US-2.1	As a citizen, I want to take a live photo inside the app, so that the report shows the real situation.	Must
US-2.2	As a citizen, I want my location filled in automatically, so that I do not have to type an address.	Must
US-2.3	As a citizen, I want the description to be optional, so that I can report quickly.	Should
US-2.4	As a citizen, I want to log in only when I am ready to submit, so that I can first explore the app.	Must
US-2.5	As a citizen, I do not want the photo saved in my phone, so that my storage and privacy are not affected.	Must
US-2.6	As a citizen, I want my complaint to submit automatically once the network returns, so that a weak connection does not lose my report.	Should
US-2.7	As the system owner, I want each user limited to three complaints per 24 hours, so that fake or spam reports are reduced.	Must
US-2.8	As a citizen, I want to track my complaint (submitted, evaluated, resolved), so that I know its progress.	Must

Acceptance Criteria

Given	When	Then
A citizen opens the report screen	The camera is used	A live photo is captured and shown for confirmation
Location permission is granted	The photo is taken	GPS coordinates are locked in automatically
Location permission is denied	Report is attempted	The system explains why location is required and does not submit
No description is typed	Report is submitted	The complaint is accepted
A guest has captured a photo	Submit is pressed	The user is asked to log in or register, and the draft is kept
A photo is captured	The report is completed	The image is not written to the device gallery
The device is offline	Submit is pressed	The complaint is queued and sent automatically when the network returns
A user has submitted three complaints in 24 hours	A fourth is attempted	It is blocked with a message showing when the user can report again
A complaint exists	The citizen opens My Complaints	The current status and its history are displayed

EP-3: Geofenced Routing and Councillor Management

Epic Goal: Automatically route each complaint to the right ward and give Councillors the tools to verify, assign and close cases.

Business Value: Removes manual sorting, avoids misrouted complaints and gives clear accountability at ward level.

Priority: Must

User Stories

ID	User Story	Priority
US-3.1	As the system, I want to match a complaint's GPS point against ward boundaries, so that it reaches the correct Councillor.	Must
US-3.2	As a Councillor, I want to see only reports from my ward, so that I focus on my responsibility.	Must
US-3.3	As a Councillor, I want duplicate reports from the same location grouped, so that I do not handle the same problem twice.	Should
US-3.4	As a Councillor, I want urgent and long-unresolved problems shown first, so that I act on the worst cases early.	Should
US-3.5	As a Councillor, I want to view the citizen's photo, so that I can verify the complaint is genuine.	Must
US-3.6	As a Councillor, I want to assign a task to a ward worker easily, so that cleanup starts quickly.	Must
US-3.7	As a Councillor, I want to be notified when work is completed or delayed, so that I can follow up.	Should
US-3.8	As a Councillor, I want to compare Before and After photos side by side and close the case myself, so that no case is closed without my approval.	Must

Acceptance Criteria

Given	When	Then
A complaint with GPS coordinates is submitted	Routing runs	It appears on the dashboard of the Councillor whose ward polygon contains the point
A point falls outside every ward	Routing runs	The complaint is flagged for Super Admin review
A Councillor opens the dashboard	The list loads	Only complaints from that Councillor's ward are shown
Several reports come from nearly the same spot	The list loads	They are grouped into one case with a report count
Cases have different ages and urgency	The list loads	Urgent and long-unresolved cases appear at the top
A Councillor opens a complaint	Details are viewed	The citizen photo, location and description are displayed
A Councillor selects a Worker	Assign is confirmed	The task appears in the Worker's list and status becomes Assigned
A Worker uploads an After photo	The Councillor opens the case	Before and After photos are shown side by side
A Councillor approves the comparison	Close is pressed	The status becomes Closed and the citizen is notified

EP-4: Worker Task Execution and Proof of Work

Epic Goal: Give ward workers a light, fast mobile view to receive tasks, reach the location and prove the cleanup was done.

Business Value: Makes cleanup measurable and prevents unverified or fake completion.

Priority: Must

User Stories

ID	User Story	Priority
US-4.1	As a Worker, I want my assigned tasks to appear instantly on my phone, so that I can start without delay.	Must
US-4.2	As a Worker, I want to see the task location and the citizen's photo, so that I know what to clean.	Must
US-4.3	As a Worker, I want a map view with exact location and directions, so that I can reach the site.	Should
US-4.4	As a Worker, I want to mark Start Work on arrival, so that my start time is recorded.	Must
US-4.5	As a Worker, I want to capture a timestamped, geo-tagged completion photo, so that I can prove the work.	Must
US-4.6	As a Worker, I want a reminder if my proof upload is delayed, so that I do not forget it.	Should
US-4.7	As a Worker, I want to mark start and take photos offline, so that poor network does not stop my work.	Should
US-4.8	As a Worker, I want to see my completed task history, so that I can review my past work.	Could

Acceptance Criteria

Given	When	Then
A Councillor assigns a task	The Worker opens the app	The task is listed with location and citizen photo
A Worker opens a task	Map is selected	The exact location and directions are displayed
A Worker is at the site	Start Work is pressed	Status becomes In Progress and the start time is recorded
The Worker finishes the cleanup	A proof photo is taken	The photo is stored with timestamp and GPS coordinates
The proof photo location is far from the complaint location	Upload is attempted	The system warns the Worker and flags the proof for the Councillor
Proof is not uploaded within the allowed time	The reminder runs	The Worker receives a reminder and the Councillor is told of the delay
The Worker is offline	Start or photo capture is done	The data is saved locally and synced once online
A Worker opens history	The list loads	Only that Worker's completed tasks are shown

EP-5: Hotspot Detection, Escalation and Alerts

Epic Goal: Detect areas with significant complaints, escalate them to the Zonal Executive Magistrate and warn nearby citizens.

Business Value: Brings high-risk areas to senior attention without requiring the Magistrate to log in, and helps citizens avoid risk.

Priority: Must

User Stories

ID	User Story	Priority
US-5.1	As the system, I want to flag an area as a hotspot when complaints cross a threshold, so that serious clusters are noticed.	Must
US-5.2	As the system, I want to generate a 48-hour secure link and email it to the Zonal Executive Magistrate, so that escalation is instant.	Must
US-5.3	As a Magistrate, I want to open the link without logging in, so that I can view the situation immediately.	Must
US-5.4	As a citizen, I want a notification when I enter within 200 m of a high-risk hotspot, so that I can take care.	Should
US-5.5	As a Super Admin, I want to configure the complaint threshold and the Magistrate email per zone, so that escalation matches local needs.	Should

Acceptance Criteria

Given	When	Then
An area's complaints reach the threshold	The check runs	The area is flagged as a Hotspot on the maps
A hotspot is created	Escalation runs	A secure link valid for 48 hours is emailed to the Magistrate
The email cannot be sent	Escalation runs	The failure is logged and retried, and the Super Admin is alerted
The Magistrate opens a valid link	The page loads	A read-only report with map, photos and counts is shown without login
The link is older than 48 hours or altered	It is opened	Access is denied and an expiry message is displayed
A citizen with alerts enabled enters 200 m of a hotspot	The location is checked	A push notification is delivered
A Super Admin changes the threshold	Settings are saved	The new value applies to future checks

EP-6: System Administration and Reporting

Epic Goal: Let the City Corporation manage users, wards and settings and monitor activity across the city.

Business Value: Provides oversight, supports planning and keeps an audit trail of important actions.

Priority: Should

User Stories

ID	User Story	Priority
US-6.1	As a Super Admin, I want to manage ward boundaries, so that routing remains accurate.	Must
US-6.2	As a Super Admin, I want to disable or reactivate accounts, so that misuse can be stopped.	Must
US-6.3	As a Super Admin, I want a city-wide dashboard of complaints, hotspots and resolution times, so that I can plan.	Should
US-6.4	As a Super Admin, I want to view audit logs of key actions, so that actions are traceable.	Should

Acceptance Criteria

Given	When	Then
A ward polygon is added or edited	It is saved	New complaints use the updated boundary
A Super Admin disables an account	The user next attempts to log in	Access is denied
A Super Admin opens the dashboard	Filters are applied	Counts, hotspots and average resolution time are shown
A user performs a key action (assign, close, role change)	The action completes	An audit log entry is created with user and time

5. System Workflow

The end-to-end flow follows five phases. It reflects the project flow chart with the updated email-based escalation.

Phase	Step	Description
1. Citizen Reporting	1-3	Citizen opens the report screen, captures a live photo with GPS lock, adds an optional description and submits.
2. System Routing and Escalation	4	The report is routed to the Councillor dashboard by geofencing. If the area has significant complaints, the system sends a 48-hour link by **email** to the Magistrate, flags a Hotspot and alerts citizens within 200 m.
3. Validation and Assignment	5-6	The Councillor inspects the photo and assigns a cleanup task to a Worker.
4. Cleanup and Proof	7-8	The Worker travels to the site, resolves the hazard and uploads a live photo with timestamp and GPS.
5. Final Verification	9-10	The Councillor compares Before and After photos and closes the task after official approval.

Flow summary: Start, Citizen Reporting, System Routing, Significant complaints? (Yes: Escalation by email, Hotspot flag, 200 m alert) (No: continue), Validation and Assignment, Cleanup and Proof, Final Verification, End.

6. Non-Functional Requirements

6.1 Performance Requirements

- Pages and API calls shall respond within 3 seconds under normal load.
- The report screen shall be usable on a slow mobile connection, with images compressed before upload.
- Ward matching shall use a spatial index so that routing completes within 2 seconds.
- The system shall handle simultaneous complaint bursts without losing or duplicating submissions.

6.2 Security Requirements

- Passwords shall be stored using a strong one-way hash.
- Four-tier RBAC shall be enforced on the server for every request.
- All traffic shall use HTTPS.
- Magistrate links shall use random, signed tokens that expire after 48 hours and expose no other data.
- A Councillor shall access only the complaints of the assigned ward.
- Rate limiting shall enforce the limit of three complaints per user per 24 hours and protect login from brute force.
- Uploaded files shall be validated for type and size.
- Important actions shall be written to an audit log.

6.3 Reliability and Availability Requirements

- The system should be available 24/7 apart from planned maintenance.
- Complaints queued offline shall not be lost and shall sync once only (no duplicates).

- Regular database backups shall be taken.
- Failed email or upload operations shall be retried and shall not corrupt existing data.

6.4 Usability Requirements

- The interface shall be fully available in Bangla and English.
- A citizen shall be able to submit a report in no more than four steps after opening the report screen.
- The landing view shall explain the app before asking for login.
- Only name and email shall be requested at sign-up.
- Error messages shall be simple and suggest how to fix the problem.
- The Worker app shall stay light and fast on low-end phones.

6.5 Privacy Requirements

- Photos taken in the app shall not be stored on the user's device gallery.
- Only the minimum personal data (name and email) shall be collected.
- Location shall be collected only at the time of reporting or for hotspot proximity alerts, with user permission.

6.6 Maintainability and Portability Requirements

- The code shall be modular, separating routes, services and data access.
- External services (storage, email, maps) shall sit behind service layers so they can be replaced without major rewrites.
- The PWA shall work across major browsers and screen sizes.

6.7 Data Integrity Requirements

- Each complaint shall keep its original GPS coordinates and photo unchanged.
- Status changes shall keep a history with user and time.
- A complaint shall not be closed without a Worker proof photo and Councillor approval.
- Duplicate accounts with the same email shall be prevented.

7. Database Requirements

The system uses PostgreSQL with PostGIS. The main data to be stored is listed below.

Entity	Key Data
User	name, email, password hash, role, ward (if any), language, account status, verified flag
Ward	name, boundary polygon (PostGIS geometry), zone, Councillor
Zone	name, Magistrate email, complaint threshold
Complaint	citizen, location point, Before image URL, description (optional), ward, status, priority, created time, duplicate group
Complaint Status History	complaint, old status, new status, changed by, time
Task Assignment	complaint, Worker, assigned by, start time, assigned time, due time
Proof of Work	task, After image URL, location point, timestamp, verified result
Hotspot	area geometry, complaint count, flagged time, status
Escalation Link	hotspot, token hash, created time, expiry time (48 hours), email delivery status
Notification	user, type (reminder, hotspot, status), message, sent time, read flag
Submission Counter	user, window start, count (for the 3-in-24-hours rule)
Audit Log	user, action, target, time

Images are held on Cloudinary and only their URLs and public IDs are stored in the database.

8. Technology Stack

Layer	Technology
Front end	React.js PWA, Tailwind CSS
Back end	Node.js (Express)
Database	PostgreSQL with PostGIS
Media storage	Cloudinary (replaces Firebase Cloud Storage)
Maps	Mapbox / Google Maps
Notifications	Email service for escalation and account mail (replaces Twilio / SSL Wireless SMS); browser push for alerts
Tools	Visual Studio Code, Git and GitHub

9. Proposed Timeline

Activity	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Environment Setup & DB Design	■												
Backend API Construction		■	■										
Frontend UI (React PWA)			■	■									
System Integration				■	■								
Dashboard Skeleton & Prep					■	■							
Progress Presentation						■							
Media Handling & Cloud Storage (Cloudinary)							■	■					
Spatial DB & Mapping Integration								■	■				
Cleanup Verification Workflow									■	■			
Threshold Automation & Email Escalation										■	■		
Location-Based Proximity Alerts											■	■	
Quality Assurance & Polish												■	
Final Project Presentation													■

The timeline is carried forward from the approved proposal. The SMS alert activity is now covered by email escalation.

10. Definition of Done

A user story is considered complete when:

- The functionality is implemented as described in the user story.
- All acceptance criteria are satisfied.
- Code is reviewed and merged into the main branch through Git/GitHub.
- Unit and integration tests pass.
- Role-based access and authentication are verified for the feature.
- Location, photo and personal data handling follow the privacy requirements.
- The feature works on a mobile browser and in both Bangla and English.
- No critical bugs remain open.
- Related documentation is updated.
- The feature is demonstrated to the supervisor and accepted.

11. Glossary

Term	Meaning
PWA	Progressive Web App; a website that behaves like an installable mobile app
PostGIS	Spatial extension of PostgreSQL used for location and boundary queries
Geofencing	Matching a GPS point to a defined area, here a ward boundary or a 200 m radius
Hotspot	An area flagged because complaints crossed the configured threshold
RBAC	Role-Based Access Control; permissions depend on the user's role
Ward	The smallest administrative division handled by a Ward Councillor
Zonal Executive Magistrate	Senior official who receives the hotspot escalation link
MoSCoW	Must, Should, Could, Won't; a method for prioritising requirements